Numerical Simulation of Quantized Vortices with Small Core Size via Vortex Tracking

Thomas Ackermann

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1 Definitions

Ginzburg-Landau Hamiltonian with core size ε: $E_\varepsilon(u) = \int_\Omega \frac{1}{2}|\nabla u|^2 + \frac{1}{4\varepsilon^2}(1-|u|^2)^2$
Renormalized energy: $W(a,d) = -\pi \sum_{1\le i\ne j\le N} d_i d_j \log|a_i - a_j| - \pi \sum_j d_j R(a_j; a, d)$
Boundary terms in the renormalized energy $R(x; a, d)$ which solves

$$\Delta R = 0, \text{ in } \Omega, \quad R = -\sum_{j=1}^{N} d_j \log|x - a_j|, \text{ on } \partial\Omega \tag{1}$$

Supercurrents $j(\psi) = \frac{1}{2i}(\overline{\psi}\nabla - \psi\nabla\overline{\psi})$
Vorticity: $J\psi = \frac{1}{2}\nabla \times j(\psi)$

2 Analytical results of $\varepsilon \to 0$

In the singular limit $\varepsilon \to 0$, the vorticity concentrates as $J\psi_\varepsilon(t) \;\to\; \pi \sum_{j=1}^{N} d_j\, \delta_{a_j(t)}$, and the vortex positions $a(t) = (a_1(t), \dots, a_N(t))$ solve the *point vortex system*

$$\dot{a}_j(t) = -\frac{1}{\pi}\, d_j\, J_{\text{symp}}\, \nabla_{a_j} W(a(t), d), \qquad a_j(0) = a_j^0, \tag{2}$$

where $J_{\text{symp}} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$.

3 Vortex tracking

3.1 Computing R

The solution R is harmonic Harmonic Polynomials are used to solve the PDE. These are given as $h_1 = 1$, $h_{2m} = \Re(x+iy)^m$ and $h_{2m+1} = \Im(x+iy)^m$ for $(x,y) \in \mathbb{R}^2$.
R can be solved numerically like this:

1. Choose a maximum degree n and fix it. Compute $(\mathbb{P}_n g)(\exp(i\theta)) = \sum_{k=-n}^{n} \hat{g}(k \exp(ik\theta))$ with $\hat{g}(k) = \frac{1}{2\pi}\int_0^{2\pi} g(\exp(i\theta)) \exp(-ik\theta)\, \mathrm{d}\theta$
2. Compute the Fourier coefficients of the Dirichlet boundary condition up to order n for instance using a FFT

3. Compute the approximate solution R_n as the harmonic expansion of $\mathbb{P}_n g_a$

$$R_n(r\exp(i\theta)) = \sum_{k=-n}^{n} r^{|k|}\hat{g}_a(k)\exp(ik\theta) \tag{3}$$

3.2 Method

Step 1: Constructing well-prepared initial conditions Define initial vortex positions and degrees. Set up the initial phase H such that the Neumann boundary condition is satisfied. Compute the radial function f_ε.
Step 2: evolve vortex positions via Hamiltonian ODE solve the ODE using RK4 and compute R.
Step 3: Smoothing At time $t > 0$ build back an approximation of the wave function from the vortex positions $a(t)$ as

$$u^*(x; a(t), d) = \exp(iH(x))\prod_{j=1}^{N}\left(\frac{x-a_j}{|x-a_j|}\right)^{d_j} \tag{4}$$

Smoothing is then applied to get the approximation of the wave function at time t as:

$$\psi_\varepsilon^*(x; a^0, d) = u^*(x; a^0, d)\prod_{j=1}^{N} f_\varepsilon(|x-a_j^0|), \quad x \in \overline{\Omega} \tag{5}$$

Key insight: ϵ only enters through the 1D preprocessing. It is no longer a discretization bottleneck.

Lemma 3.1. *Let $\Omega \subset \mathbb{R}^2$ be the unit disk, let $\{a(t)\}_{t\geq 0}$ denote the exact solution to the Hamiltonian mechanics. Let $b(t)$ be the approximated ODE trajectory with the same degrees and initial positions b^0 with harmonic polynomials of degree up to n and time step δ_t. Suppose that for some $T > 0$,*

$$\rho_T = \min_{t\leq T}\min_{j\neq k}\{|b_j(t)-b_k(t)|, dist(b_k(t),\partial\Omega), |a_j(t)-a_k(t)|, dist(a_j(t),\partial\Omega)\} > 0 \tag{6}$$

is such that $0 < \rho_T < 1$. Then we have:

$$|a(t)-b(t)| \leq C(|a^0-b^0| + (1-\rho_T)^n n^{1-l}) + C\delta t^4 \tag{7}$$

References

[Car+25] Thiago Carvalho Corso et al. "Numerical simulation of the Gross-Pitaevskii equation via vortex tracking". In: *Mathematics of Computation* 95.357 (Feb. 2025), pp. 227–262. ISSN: 0025-5718. DOI: 10.1090/mcom/4071. URL: http://dx.doi.org/10.1090/mcom/4071.